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Association Between Sports Injuries and Sleep as Measured by Sleep Actigraphy: A Scoping Review

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Association Between Sports Injuries and Sleep as Measured by Sleep Actigraphy: A Scoping Review

Abstract

Introduction

The purpose of this scoping review was to determine if there is an association between sleep patterns, as measured by sleep actigraphy, and athletic injuries and performance.

Methods

Studies were included from Pubmed, Embase, and SPORTDiscus if they measured sleep using actigraphy, evaluated sports injuries and athletic performance. Systematic reviews, narrative reviews and those not using actigraphy or which focused on concussions instead of other sports injuries were excluded.

Results

A total of 17 studies met inclusion criteria, examining athletes from 10 different sports. Of the seven studies evaluating the relationship between sleep metrics (sleep latency, efficiency, duration, wake after sleep onset) and injury occurrence, four studies (57%) found that decreased sleep correlated with increased injury occurrence in elite soccer, paralympic and adolescent track and field athletes. Eight studies (47%) evaluated the relationship between sleep metrics and performance (reaction time, sports specific tasks, aerobic and anaerobic capacity), finding that decreased sleep metrics correlated with decreased aerobic and anaerobic performance in rugby athletes, cyclists, triathletes, Mixed Martial Arts, and Dutch elite youth athletes.

Conclusion

Findings of this study show that there is no consensus on the relationship between the amount or quality of sleep and rates of sports-related injuries in adult athletes. Poor sleep metrics (total

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sleep time/duration, latency, efficiency) were correlated with a decrease in response time in cyclists and Dutch elite youth athletes. Worsened sleep metrics were also found to decrease aerobic performance in physically active males, MMA fighters, rugby players, triathletes, and cyclists. In summary, these findings suggest that there is an association between poor sleep and worsened athletic performance, but the evidence for injury risk is inconclusive.

Keywords:

Injury Prevention, Performance Improvement, Sleep, Actigraphy, Performance, Injury, Review

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Introduction

Sports science, a field that has grown substantially over the past decade, has focused on optimizing athlete performance and recovery. Sleep has been shown to have a major impact on both factors, with recent studies suggesting that a lack of sleep can lead to changes in both physical and mental performance for athletes.[1,2] Lack of sleep has been shown to be a risk factor for injury occurrence in adolescent athletes; however, it has not been consistently established in adults[3,4]. Evidence also suggests that poor sleep can negatively impact an athlete's performance

in aerobic exercises such as race times, endurance tests, V02 max and heart rate recovery and in sport specific accuracy and reaction time.[5-8] However, there is conflicting evidence on whether anaerobic performance on short burst exercises like the Wingate test, and sprints are equally impacted.

Prior studies have utilized a variety of methods to evaluate the amount and quality of sleep, ranging from subjective sleep diaries and questionnaires to the more invasive polysomnography.[2,9] However, there are limitations with both methods, prompting the search for a less invasive yet objective approach to measuring sleep.[9] Sleep diaries and questionnaires allow for flexibility and cost savings, making them attractive options for sleep researchers. However, when self-reported sleep reports were compared to an objective measure like actigraphy, athletes were found to over report the total time they were asleep and under report the amount of times they awoke during the night.[10]

Polysomnography, which is often considered the gold standard for sleep studies, allows for a comprehensive assessment of sleep-wake cycles, often in a laboratory setting. However, this method requires significant time and resources from both study researchers and patients as it involves the patient sleeping in a laboratory setting while being monitored by expensive equipment. Wrist actigraphy is a feasible and attractive option to study sleep in athletes. With advances in technology and the advent of commercially available and popular smart watches, wrist actigraphy allows for the measurement of sleep data without any cumbersome equipment or personnel. These watches measure wake-sleep cycles by monitoring the wearer's position in space with a built-in accelerometer.[2,9]

While many studies have focused on using subjective sleep diaries or questionnaires to determine the impact of sleep on athletic performance, the use of sleep actigraphy remains a

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relatively underexplored avenue to objectively evaluate sleep patterns and their association with athletic injuries and performance. The purpose of this scoping review was to fill this gap by using actigraphy to objectively determine if there were any associations between sleep metrics and athletic injuries and performance.

Methods

This scoping review was conducted according to PRISMA extension for scoping reviews checklist. A literature search was performed in January 2024 on electronic databases (PubMed, Medline, Embase, SPORTDiscus) using search terms such as "Wrist Actigraphy and Athlete injury", "Sleep Actigraphy and Sports Injury", "Sleep Actigraphy and Sports Performance", and "Wrist Actigraphy and Athletic Performance". The institutional health sciences library assisted in compiling relevant search terms. Additional relevant studies were identified by searching the reference lists of previously included studies.

An updated search was performed in January 2025, and no additional sources were identified. Studies were included if they evaluated male or female athletes of any age or sport who used sleep actigraphy to monitor the quantity and quality of sleep, in order to identify its relationship to the incidence of athletic injuries or performance. Studies evaluating concussions were excluded due to their distinct pathophysiology from musculoskeletal injuries and the increased variability they introduce when comparing injury occurrence between contact and non-contact sports. Additionally, studies had to be peer reviewed articles in the English language published within the past 24 years. Abstracts, commentaries, systematic reviews, or any other articles that were not peer-reviewed studies were excluded. Studies monitoring the use of sleep with subjective questionnaires or any measure other than actigraphy were also excluded. Athletic

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performance was not categorized the same way in each study, some focused on time trial and race performances, others VO2 max, heart rate recovery, sprints, vertical jump, response time, and sport specific accuracy. Sleep metrics in this review are as follow: total sleep time or duration, sleep latency or sleep onset latency (defined as how long it takes someone to fall asleep after getting into bed), wake after sleep onset (WASO) (how many minutes someone is awake after they've initially fallen asleep) and sleep efficiency (of the amount of time a person has been in bed, how much of that time have they been asleep).

Consistent with scoping review methodology, a formal risk of bias or study quality assessment was not performed, as the primary aim of this review was to map the existing literature and identify knowledge gaps rather than to evaluate methodological rigor.

Study Selection Criteria and Procedures

Abstracts identified during the initial database search were screened by at least two authors to determine whether the studies met all inclusion criteria. The full text was subsequently reviewed for all studies that met the inclusion criteria after abstract review. Any discrepancies were resolved by a third author.

Summary Measures

Data items extracted from each study included participant demographics such as age, sex, and which sport they participated in, as well as the statistical significance of sleep metrics effects on injury prevalence and athletic performance. Statistical analysis and significance, as measured by P-values, were included in the scoping review.

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Results

In total, 237 studies were identified in the initial search. Two hundred and twenty studies were excluded due to being duplicates or out of the scope of this study, leaving a total of 17 studies which met the inclusion criteria. These studies included 455 athletes, ranging from adolescent, to college, and professional/elite athletes, (23% female, n=103; 77%, male n=352) from a variety of sports; cycling, triathlon, rugby, soccer, football, track and field, rowing, tennis, Paralympics, and mixed martial arts (Table 1). Studies included evaluated the association between injury rate/occurrence or performance and sleep metrics.

The studies identified demonstrated mixed evidence of the association between sleep and injury rate (Table 2). Of the seven studies which evaluated the association between sleep metrics and injury rate or occurrence, four (57%) found a negative correlation between sleep metrics (eg. Sleep Onset Latency (SOL), Wake After Sleep Onset (WASO), Sleep Efficiency) and injury occurrence.[11-14] Silva et al. demonstrated an association between sleep onset latency and injuries in Paralympic athletes.[11] Viegas et al. found that in adolescent track and field athletes, WASO was an associated predictor for having had an injury in the last 6 months.[12] In a study examining injury rates among elite soccer athletes, Silva et al. found that sleep efficiency (amount of time asleep out of time spent in bed) was associated with an increased amount of injuries, injury severity, and the amount of time absent from play.[13] The authors found that longer WASO also had a significant association with an increased amount of injuries among soccer players. In 2019, Nedelec et al., described a significant association between injury occurrence and worse SOL and Sleep Efficiency in an elite soccer athlete both the night before the injury and the week leading up to the event.[14]

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The remaining three studies (43%) found no association between sleep duration and injury occurrence or rate.[15-17] Patel et al. evaluated high school track and field athletes and found no significant difference between total sleep time of the athletes that suffered an injury compared to those who did not.[15] Burke et al. found no evidence that sleep duration had a significant association with injuries in collegiate football players.[16] Dennis et al. evaluated whether sleep duration and sleep efficiency had an association with injury incidence in athletes in the Australian Football League.[17] Once a player was injured, sleep for the week of the injury was compared to the average of the two weeks previous. The authors found no significant difference in sleep duration or efficiency in the week of an injury compared to the prior two weeks.[17]

Sleep was found to be associated with athletic performance (either physical or technical); however, the type of performance (aerobic or anaerobic) impacted varied. All eight studies that evaluated sleep metrics (eg. sleep duration, sleep efficiency, sleep latency) and athletic performance (100%) found an association between sleep metrics and a decrease in athletic performance (Table 2).[18-25] In these studies, athletic performance was measured by intermittent sprint performance, VO_2 max, heart rate recovery, 15 meter timed weighted sled push, vertical jump, triathlon race times, cycling race times, box drop maximal vertical jump and sports specific metrics such as accuracy.[18-25] However, not all of the studies demonstrated correlations between sleep and all aspects of athletic performance. For example, Vitale et al. demonstrated that acute sleep restriction (≤5 hrs) was associated with decreased shot accuracy in junior tennis players, but was not associated with athletic performance in repeated sprint tests.[23] Knufinke et al. found no association between sleep and sports specific performance in elite Dutch youth athletes, but did demonstrate a negative correlation between sleep quantity and reaction time.[25] The authors also found that shorter sleep onset latency improved gross motor ability such as

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vertical jump, twenty-meter sprints, and constant power cycling tests. Teece et al. demonstrated that, longer sleep duration was associated with increased aerobic capacity and improved body composition in professional rugby players, but did not find an association between sleep duration and performance in measurements of strength or speed such as squat, bench, chin ups, or five and ten meter sprints.[24]

Additionally, in 2020, Baati et al. found an association between lack of sleep and a decrease in distance perception in male college soccer players.[26] Kolling et al. found a positive correlation between sleep time and recovery rate in junior rowers, and Mah et al. found a negative correlation between sleep quantity and reaction time, as well as an association between sleep restriction and an increase in joint movement variability in professional male cyclists.[22,27] All of these factors, although not measured in the other studies, could play a role in an athletes' game day performance.

Overall, the relationship between poor sleep and injury risk was inconclusive, whereas all of the studies in this review found that as sleep worsens so do elements of an athlete's performance. Performance in time trials, race times, vertical leap, tests of endurance with repeated sprints and sled push, as well as response time, depth perception, recovery, VO2 max, heart rate recovery and body composition were all shown to worsen when athletes have poor sleep, while an athlete's performance on sprints and sport specific accuracy appeared to be inconclusive.[18-25]

Discussion

The purpose of this scoping review was to determine if there is any association between sleep patterns and athletic injuries and performance using strictly actigraphy, an objective

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measure. The findings of this study suggest an association between sleep and reaction time and aerobic performance, although there is conflicting evidence about the extent that anaerobic performance (e.g. strength or speed) is affected. The studies identified demonstrated variability in terms of a possible correlation between sleep and injury rate using actigraphy.

Similar to the findings of this study, outside of actigraphy studies, the association between poor sleep and injuries in adult athletes is unclear. In a 2021 systematic review, Dobrosielski et al. explored the association between poor sleep and injuries in adult athletes.[28] The authors reviewed 12 prospective cohort studies and determined that there was insufficient evidence to conclude that poor sleep put adult athletes at increased risk of injury. Most of the evidence identifying poor sleep as a risk factor for injury comes from studies of adolescent athletes. Our own review found inconclusive evidence that adolescent athletes who have poor sleep were at increased risk for injury, Viegas et al. finding a correlation while Patel et al did not.[12,15] In contrast, a systematic review and meta-analysis by Gao et al of 7 prospective cohort studies found that there was an increased likelihood of injury in adolescent athletes who had chronically poor sleep. Of note, all 7 of the studies included in the review used self-reported questionnaires to measure sleep.[3] Further studies are necessary to determine if using actigraphy will provide similar results that consistently affirm the association between chronic sleep deprivation and increased injury rates among adolescent athletes. Even if these findings are affirmed by objective measures, adolescent athletes should not be used as a reliable analog for adult athletes as they have differing circadian rhythms than adults and greater sleep need for maturation and growth.[29]

Previous studies support the claim that sleep restriction has a negative impact on aerobic performance. Charest et al. conducted a narrative review that examined how sleep impacted

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various areas of athletes' performance and health.[5] The authors found that as a result of lack of sleep, athletes' aerobic performance (eg. cycling ergometer test) and VO2 max decreased, and their heart rate and respiratory rate increased. In a narrative review evaluating athletes' sleep disturbances and their impacts on performance, Watson demonstrated that performance on endurance tests decreased with decreased amounts of sleep.[6] Similarly, Day et al. performed a narrative review examining sleep disorders, treatments and the crucial nature of sleep in the health and performance of elite athletes from various sports.[8] The authors found that athletes with sleep deficits reported fatigue earlier in endurance tests and an increased perception of effort. These results are consistent with findings of the present study, which showed a decrease in race times for athletes such as cyclists and triathletes, and an improvement in metrics such as V02 max and heart rate seen in other athletes as their sleep duration increased.

There is mixed evidence regarding whether anaerobic performance (e.g. strength and sprinting) is affected by decreased sleep. Watson demonstrated that a clear association cannot be drawn between decreased sleep and worsened anaerobic performance.[6] This is because the results of the study showed that decreased sleep had conflicting impacts on Wingate test and sprint performance. In contrast, a narrative review by Charest et al. found that decreased anaerobic performance was associated with sleep restrictions.[5] They noted that sleep deprivation decreased athletes' peak and mean power output in the Wingate test. Aerobic and anaerobic performance may be affected by sleep restriction in different ways. Charest et al. explained that low amounts of sleep decreased muscle glycogen stores as well as increase the perception of effort, which is posited as an explanation for why the impact of sleep deprivation is more apparent in long-duration activities like endurance running than in short-term maximal-effort activities like sprinting or weight-lifting.[5]

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Other areas of performance that are impacted by sleep are psychomotor abilities such as reaction time, as well as accuracy in sports specific tasks. Day et al. evaluated the unique factors that impact athletes' sleep, treatment plans, as well as how these sleep issues affected mental and physical performance.[8] The authors found that sleep extension led to increased three point shot accuracy in basketball players, and sleep deprivation led to decreased serve accuracy among dart throwers, and tennis players' serves. Similarly, Wilkes et al. examined the relationship between sleep disorders and athletes' physiological and functional performance and found that sleep extension improves basketball players free throw and three point field goal percentages, as well as improves tennis players serving accuracy.[7] The authors also found that both soccer and basketball players suffering from sleep restriction had decreased reaction time during a Psychomotor vigilance test. Different from previous reviews, this review used strictly wrist actigraphy to measure athletes' sleep. Though questionnaires such as the Pittsburgh Sleep Quality Index (PSQI) have been clinically validated in multiple populations, recent studies suggest diminished retest reliability in assessing sleep quality of athletes over a period of 1 month.[30,31] Newer questionnaires, such as the Athlete Sleep Screening Questionnaire and Athlete Sleep Behavior Questionnaire, allow for longitudinal study in athletes without additional costs.[32,33] However, further exploration is needed utilizing these newer athlete-focused questionnaires as studies suggest slight discordance in the prevalence of sleep disturbances in athletes, depending on which questionnaire was used. In 2018, Bender et al. evaluated sleep quality in 199 Canadian national team athletes using the Athlete Sleep Screening Questionnaire.[34] The authors found that 25% of the athletes had a clinically relevant sleep problem, which was in contrast to previously reported rates of 40-50% in athletes when evaluating sleep quality with the PSQI.[34-36] In a 2025 study by Devrim et al., comparing self-

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reported sleep to actigraphy they found that their athletes overestimated the amount of time they were asleep and underestimated how much they woke during the night.[10] Actigraphy as a measure of sleep also has its limitations. Although it has been validated as a measure for chronic sleep disorders compared to polysomnography, a meta-analysis found that it may overestimate total sleep time in patients with chronic conditions when compared to polysomnography.[37,38] Chinoy et al. showed that the accuracy may vary depending on the commercial model.[39] While actigraphy is not as accurate as the gold standard, polysomnography, it may offer a more convenient way for athletic trainers to consistently track their athletes' sleep which is more accurate than self-reported questionnaires.

Limitations

There are several limitations to this study. First, although the studies included in this review have internal validity, they have highly specific participant populations, which can only be representative of their own individual sports. This unique participant pool combined with small sample sizes makes it difficult to show a trend that can be applied across multiple populations of athletes. Along with a difference in types of athletes, the studies had different designs and did not all track the same sleep metrics or use the same measures of performance. While there was some overlap, this variability makes generalized claims less reliable. Also of note, only 7 studies (of which one was a case study) explored whether sleep played a role in injuries. With such a small number of studies, it is challenging to make generalizable conclusions.

Finally, women only made up a total of 23% of the athletes in the studies included in this review. Additional research is needed to determine the potential impact of poor sleep on women

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athletes in terms of risk of injuries and overall athletic performance. It should also be noted that men and women are likely to suffer different types of injuries, and hormonal changes throughout the menstrual cycle may put women at an increased risk of injury.[40,41] Given these sex-related differences, recommendations based on data from a predominantly male population may not be completely applicable for women athletes.

Conclusion

No significant relationship was found between the amount or quality of sleep and rates of sports-related injuries in adult athletes. Worse sleep metrics (total sleep time/sleep duration, sleep latency, sleep efficiency) were correlated with a decrease in response time in cyclists and Dutch elite youth athletes, and aerobic performance in physically active males, MMA fighters, rugby players, triathletes, and cyclists. Increases in sleep latency and decreases in sleep duration also negatively impacted anaerobic performance (max vertical jump, prowler push, cycling power assessments) in MMA fighters, cyclists, Dutch elite youth athletes. Actigraphy could be a convenient and reliable measure for athletic trainers to longitudinally track an athletes' sleep in order to optimize their performance. While further longitudinal studies using standardized actigraphy are necessary to more clearly define what role sleep plays in sports-related injuries and athletic performance, athletes should be encouraged to prioritize high quality sleeping habits for healthy and successful competition.

Disclosure Statements

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Table 1 Study Demographics

Study	Population	Sample size	Female	Male	Age in years (mean ± SD)
Patel et al., 2020[15]	High school track and field athletes	17	10	7	15.9
Baati et al., 2020[26]	College soccer players	10	0	10	22.8 ± 1.3
Burke et al., 2020[16]	College football athletes	94	0	94	19.6 ± 1.7
Silva et al., 2022[11]	Paralympic athletes	20	3	17	33.65 ± 8.35
Viegas et al., 2022[12]	Track and field adolescent athletes	19	6	13	16.89 ± 2.75
Fowler et al., 2014[18]	Physically active males	10	0	10	23.9
Silva et al., 2020[13]	Elite soccer athletes	23	0	23	26.5 ± 5.2
Nedelec et al., 2019[14]	Elite soccer athlete	1	0	1	31
Peacock et al., 2018[19]	Professional MMA (Mixed martial arts) athletes	8	0	8	27.7 ± 3.4
Dennis et al., 2016[17]	Australian Footballer	22	0	22	23.8 ± 3.2
Teece et al., 2021[24]	Rugby athletes	29	0	29	23 ± 3
Kisiolek et al., 202120	Triathlon	17	3	14	37.44
Roberts et al., 2019[21]	Cyclists	9	0	9	30 ± 6
Vitale et al., 2021[23]	Junior tennis players	12	0	12	15.4 ± 2.6
Mah et al., 2019[22]	Cyclists	11	0	11	28.8 ± 4.5
Knufinke et al., 2018[25]	Dutch elite youth athletes	98	56	42	18.8 ± 3.0
Kolling et al., 2016[27]	Junior rowers	55	25	30	17.7 ± 0.6

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Table 2 Outcomes

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Authors	Population	How is sleep measured?	What's measured in relation to sleep?	Results
			Injury	
Patel et al., 2020	High school athletes	Total sleep time	Injured athletes	p = 0.06
Burke et al., 2020	College football athletes	Sleep duration	Injury	p = 0.358 , OR = 1.01; 95% CI = 0.99–1.02
Silva et al., 2022	Paralympic athletes	Sleep latency	Injuries/illness	p = 0.04
Viegas et al., 2022	Track and field adolescent athletes	Wake after sleep onset (WASO) Time awake (TA)	Injuries in last 6 months and injury occurrence	(OR=1.144) (OR=0.974)
Silva et al., 2020	Elite soccer athletes	Sleep efficiency Sleep efficiency Sleep efficiency Wake after sleep onset (WASO)	Amount of injuries Absence time after injury Injury severity Amount of injuries	p < 0.01 p = 0.02 p < 0.01 p = 0.01
Nedelec et al., 2019	Elite soccer athlete	Sleep onset latency Sleep efficiency the night before Sleep onset latency week before Sleep efficiency week before	Injury occurrence	ES = 3.1 ES = 3.2 ES = 1.6 ES = 2.8
Dennis et al., 2016	Australian Footballer	Sleep duration Sleep efficiency	Injury incidence	p > 0.05 p > 0.05
			Athletic Performance	
Roberts et al., 2019	Cyclists	Sleep extension 4 days (SE) Normal sleep (NS) Sleep	Time trial	Day 3 SR slower than NS (p < 0.02) Day 4 SE faster than NS (p < 0.02)

		restriction 4 days (SR)		
Kisiolek et al., 2021	Triathlon	Total sleep time	Race performance: Stage 1: 10-km swim 146-km cycle Stage 2: 276-km cycle Stage 3: 84.4-km run	 p = 0.019 p > 0.05 p = 0.035
Peacock et al., 2018	Professional MMA (Mixed martial arts) athletes	Sleep latency Sleep latency Sleep latency Sleep latency Sleep latency Sleep efficiency	Vertical Jump Prowler Push VO2max Heart Rate Recovery Missed session due to injuries Heart Rate Recovery	p = 0.02 p = 0.024 p = 0.006 p = 0.036 p = 0.035 p = 0.003
Teece et al., 2021	Rugby athletes	Sleep duration	Body composition with skin folds Aerobic via bronco test Squat Bench Chin up 5 m sprint 10 m sprint	p = 0.007 p = 0.022 p = 0.750 p = 0.284 p = 0.853 p = 0.799 p = 0.400
Fowler et al., 2014	Physically active males	Sleep duration and quality, simulating air travel	Intermittent Sprint test Yoyo intermittent recovery	p < 0.01
Mah et al., 2019	Cyclists	Sleep restriction	Maximal jump height Change of joint mechanics: Hip sagittal/knee frontal Change of joint mechanics: hip frontal/knee frontal Psychomotor Vigilance Task response time	p = 0.03 Δ15.5° p = 0.04 Δ11.0° p < 0.01 p = 0.03
Knufinke et al., 2018	Dutch elite youth athletes	Total sleep time Total sleep time Sleep latency	Response time Fine motor skill sport specific Gross motor (vertical jump, sprints, cycling power test)	p = 0.04 p = 0.10 p = 0.03

Vitale et al., 2021	Junior tennis players	Acute sleep restriction	Serve/Shot Accuracy:	
			Right	$p = 0.010$
			Left	$p = 0.014$
			Crosscourt backhand	$p = 0.003$
			Forehand	$p = 0.014$
			Repeated Sprints	$p > 0.05$
Baati et al., 2020	College soccer players	Partial sleep deprivation: Normal night sleep (NNS) Beginning of the night (SDB) End of the night (SDE)	Depth perception before and after:	
			35 m before exercise front angle 0 degrees	$p < 0.05$ NNS better than SDB, SDE
			and side angle 45 degrees	$p < 0.05$ NNS better than SDB, SDE
			35 m after exercise front angle 0 degrees	$p > 0.05$
			and side angle 45 degrees	$p < 0.05$ NNS better than SDB, SDE
			15m before exercise front angle 0 degrees	
			and side angle 45 degrees	
			15m after exercise front angle 0 degrees	
			and side angle 45 degrees	
Kolling et al., 2016	Junior rowers	Total sleep time	Overall Recovery Score	$p < 0.01$
			Negative emotional state	$p < 0.01$

Acronym Key: NNS: Normal night sleep, SDB: Sleep Deprivation at the Beginning of the night, SDE: Sleep Deprivation at the End of the night, SE: Sleep Extension, SR: Sleep Restriction,

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